Notes To Myself

Arc Lengths

L.Dodd

October 12, 2026
(v 3.2)

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CHAPTER 1

Curve Lengths

1.1. Introduction

In this short document we derive the general formula for the *arc length* S of a single-variable function $f(x)$. It demonstrates the value of a differential *secant line* and raises a few other questions at the end.

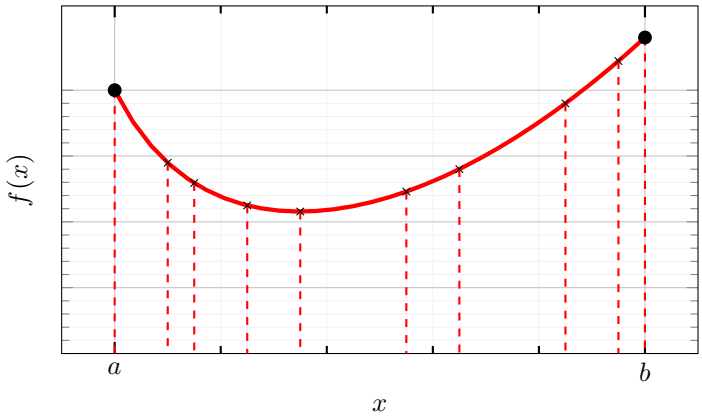
Suppose we want the length of the curve shown below. That curve is a portion of continuous function $f(x)$ over some closed interval $[a, b]$[1]



The curve is not part of a simple geometric figure so its exact value is not easily found. Finding it requires some additional mathematics.

1.2. Intervals

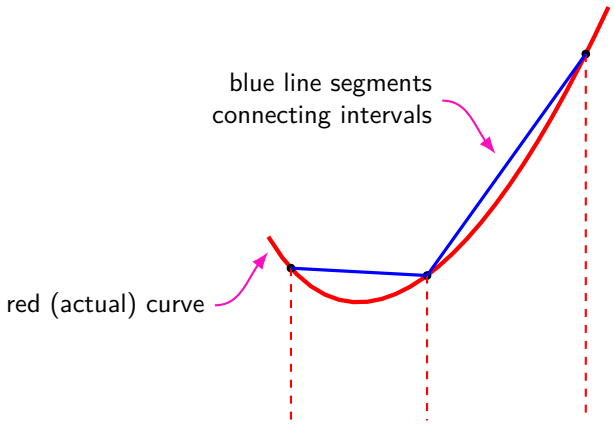
Our first step is to partition the interval $[a, b]$ into nine sub-intervals as shown



The width of the intervals is intentionally not uniform. Instead we have selected a random collection of widths.

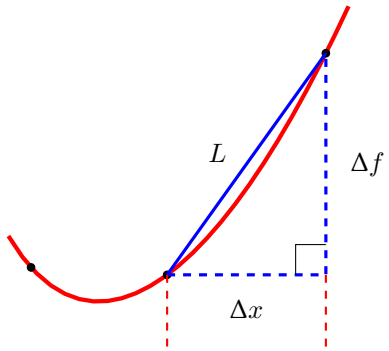
[1] The actual function is $f(x) = x^2 + 4/(2x + 1)$ with closed interval $[0, 2]$.

Within each interval we can approximate the actual curve using a line segment connecting the function at the interval end points. In other words, we will approximate using *secant lines*.[2] Here are is a close-up of a pair of adjacent intervals



We will use the lengths of the **blue** secant lines to approximate the corresponding length of the actual curve in **red**.

The length of any of the secant lines is the hypotenuse of a right triangle connecting the end points of the interval



This hypotenuse can be written conveniently in terms of the interval's width and the change in the function's value across that interval

$$\text{length of secant line for one interval:} \qquad L = \sqrt{(\Delta x)^2 + (\Delta f)^2} \qquad = \sqrt{1 + \left(\frac{\Delta f}{\Delta x}\right)^2}\,\Delta x$$

We repeat the above for all n intervals. The sum of these gives an approximation for the true arc length S

$$S \approx \sum_{i=1}^{n} \sqrt{1 + \left(\frac{\Delta f_i}{\Delta x_i}\right)^2}\,\Delta x_i$$

We use a subscript i as a simple index for the n intervals.

The above can be used to estimate S. We need only create n intervals, locate the endpoints on x and corresponding $f(x)$. It can be quite tedious, however, such summation can be shown mathematically to converge to the actual value with the introduction of more and more intervals.

1.3. Slope

We can use the **Mean Value Theorem** to re-write the summation above.

[2] A secant is defined as a line that intersects a curve at a minimum of two points

Within an any interval $[x_{i-l}, x_i]$ there exists a $c_i \in [x_{i-1}, x_i]$ such that the function's slope at that c_i will equals the secant's slope

$$\text{secant's slope} = \frac{\Delta f_i}{\Delta x_i} \;=\; \text{function's slope at } c_i \;= \frac{\mathrm{d}f}{\mathrm{d}x}\bigg|_{x=c_i} = f'(x)\bigg|_{x=c_i}$$

Thus, for each interval i in the summation we can replace the fraction with $f'(x)$ evaluated at some point c_i in that interval

$$S \approx \sum_{i=1}^{n} \sqrt{1 + [f'(c_i)]^2}\ \Delta x_i$$

where $c_i \in [x_{i-1}, x_i]$ and $\Delta x_i = x_i - x_{i-1}$.

The above is still an approximation since we are using a finite number of intervals. The advantage of this formulation is as follows. Breaking the interval $[a, b]$ into n pieces is called partitioning. Specifically

$$x_0 < x_1 < x_2 < x_3 < \ldots < x_{n-1} < x_n \qquad \text{where} \qquad x_0 = a \quad x_n = b$$

Next, we define $\|\Delta\|$ as the *norm* of the partition of the interval $[a, b]$. The norm will be largest interval within it

$$\text{norm:} \qquad \|\Delta\| \equiv \max\{|x_i - x_{i-1}| : i = 1, 2, 3, \ldots, n\}$$

In the limit as the norm goes to zero, we find the summation becomes a definite integral

$$S = \lim_{\|\Delta\| \longrightarrow 0} \sum_{i=1}^{n} \sqrt{1 + [f'(c_i)]^2}\ \Delta x_i = \int_{x=a}^{x=b} \sqrt{1 + [f'(x)]^2}\ \mathrm{d}x$$

1.4. Example

We can try to evaluate this for this function

$$f(x) = x^2 + \frac{4}{2x+1} \qquad \Longrightarrow \qquad f'(x) = 2\,x - \frac{8}{(2x+1)^2}$$

between $a = 0$ and $b = 2$. This is the example drawn. The arc length would be the value of this definite integral

$$S = \int_0^2 \sqrt{1 + \left[2\,x - \frac{8}{(2x+1)^2}\right]^2}\ \mathrm{d}x$$

The above is very messy and (as far as the author knows) can NOT be done analytically. Numerical integration gives

$$s = 5.10510357182\ldots \approx 5.1 \qquad \checkmark$$

1.5. Nuts and Bolts

The arc length can be determined analytically for surprising few functions. We can do it for a constant $f(x) = c$ from 0 to 2

$$S = \int_0^2 \sqrt{1 + (0)^2}\ \mathrm{d}x = x\Big|_0^2 = 2$$

We can do it for a constant $f(x) = x$ from 0 to 2

$$S = \int_0^2 \sqrt{1 + (1)^2}\ \mathrm{d}x = \sqrt{2}\ x\Big|_0^2 = 2\sqrt{2}$$

However, something as simple as $f(x) = 1/x$ or even $f(x) = \sqrt{x}$ can be either be extremely difficult or impossible to do analytically. For example, there is no closed-form equation for the perimeter of a simple ellipse. For this reason calculus books have very particular functions for practice. Things like $f(x) = x^{4/3}$ or $f(x) = \ln(\cos(x))$ or $f(x) = 3 + \frac{1}{2}\cosh(2x)$.

1.6. General

In general the arc length S for any continuous function $f(x)$ between points $x = a$ and $x = b$ is given by this integral

$$S \equiv \text{arc length (between } a \text{ and } b) = \int_a^b \sqrt{1 + \left(\frac{\mathrm{d}f}{\mathrm{d}x}\right)^2}\ \mathrm{d}x$$

Also, the *differential arc length* $\mathrm{d}S$ of $f(x)$ at x is given by

$$\mathrm{d}S \equiv \text{differential arc length (at } x) = \sqrt{1 + \left(\frac{\mathrm{d}f}{\mathrm{d}x}\right)^2}\ \mathrm{d}x$$

1.7. Usage

While it is easy to set up the integral to find S, evaluating that integral often is a challenge. Getting a closed-form analytical arc length S is NOT easy and mostly impossible. In nearly all cases, a numerical evaluation of the integral will be needed.

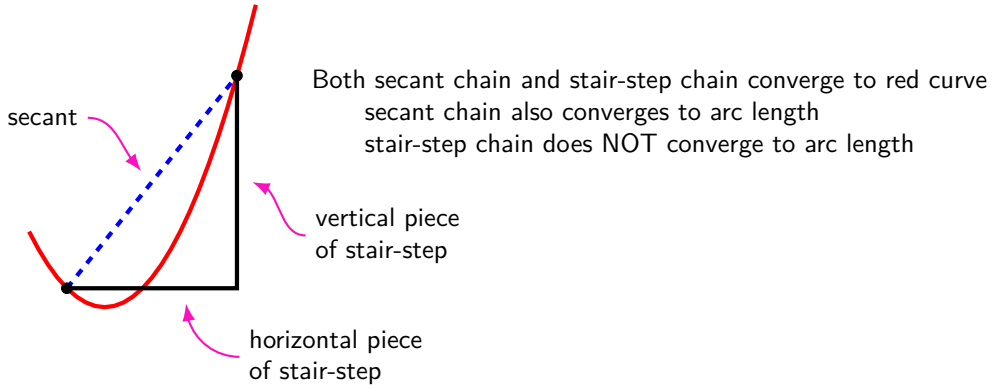
1.8. The Secant Line's Virtue

We used secant line segments for the differential element to get the integral for S. This was crucial. By using more secant line segments in the polygonal chain over the domain we get

(i) a chain that converges to the actual curve
(ii) a chain's length that converges to the curve's arc length

Naively, one would think if the first statement is true, then this would mean the second is as well. But this is not the case. Other polygonal chains converge to the curve but their length will NOT converge to the curve's arc length.

Consider a *stair-step polygonal chain*. These contain alternating horizontal and vertical segments in place of the secant segment



Like the secant, the stair-step chain does converge to the curve. Unlike the secant, the stair-step's length will NOT converge to the curve's arc length. Surprisingly, the stair-step chain length will be the same constant no matter how many intervals are used across the domain.

This is well known topic called *The Staircase Paradox*. It shows that the limits of curves do not preserve length. A curve can undergo what mathematicians call *uniform convergence* but the length of that curve is not a continuous function of the curve. Thus, the stair-step chain always has the same constant total length regardless of how many steps are used. This quite remarkable really.